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Shaking table experimental characterization of a large scale vertical isolation system

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Abstract. *Base isolation is a widely researched, developed, and utilized passive control strategy for civil and mechanical engineering structures, acclaimed for its effectiveness and reliability. Despite the extensive and successful application of horizontal isolation devices in full-scale civil engineering structures, vertical isolation remains relatively uncommon, even though the vertical seismic component is particularly hazardous in near-fault zones.*

This paper presents preliminary results from an experimental campaign aimed at dynamically characterizing an innovative Quasi-Zero Stiffness (QZS) vertical isolation system. The system tested comprises three identical QZS devices, each designed to statically support the designated payload and dynamically isolate it through an engineered combination of vertical and horizontal springs. The experiments were conducted at the Experimental Dynamic Laboratory of the L.E.D.A. Research Institute, employing stepped sine, swept sine, and natural seismic tests to determine the dynamic properties of the isolation system, varying the payload mass and the horizontal stiffness preload.

The findings, in terms of acceleration and displacement of the isolated mass, affirm the efficacy of the proposed QZS isolation system. Additionally, the device's robustness is evidenced by its proficient non-linear filtering performance in response to seismic inputs of varying amplitude, frequency content, and non-stationarity.

1 Introduction

Seismic hazard is undoubtedly one of the most important causes of damage to civil structures and, very often, it is the cause for the loss of lives and goods as recently showed during the 7.8 MW earthquake that struck southern and central Turkey and northern and western Syria, in February 2023. Seismic isolation is a method of protecting a building from major earthquakes by installing some isolation and energy absorbing devices between the building superstructure and its foundation [1]. Despite the extensive and successful application of horizontal isolation devices in full-scale civil engineering structures, vertical isolation remains relatively uncommon, even though the vertical seismic component is particularly hazardous in near-fault zones [2].

For vertical excitation, it is difficult to achieve a reliable isolation strategy due to the need to maintain sufficient stiffness and strength in the direction of gravitational loads. The need to maintain high stiffness for gravity loading while allowing flexibility for isolation during earthquakes has led to research on the use of High-Static-Low-Dynamic Stiffness Systems (HSLDSS) and in particular Quasi-Zero Stiffness Systems (QZSS), which have zero equivalent stiffness in the equilibrium position.

The idea of this research has been recently developed in [3] where the analytical model and a small-scale prototype of an HSLDSS device has been numerically and experimentally studied. As a further development of that work, an innovative vertical seismic isolation system made by three identical large-scale HSLDSS devices has been designed and manufactured. Each HSLDSS device has been designed to statically support the designated payload and dynamically isolate it through an engineered combination of vertical and horizontal springs.

The present work reports the results of a preliminary experimental campaign carried out taking advantage of the 6-DOF shaking table system of the L.E.D.A. (Laboratory of Earthquake engineering and Dynamic Analysis) Research Institute at the "Kore" University of Enna [4, 5]. The following is a summary of the paper's outline. Section 2 introduces the experimental setup and the test protocol adopted. Section 3 presents the experimental results obtained. Finally, some conclusions are drawn.

2 Experimental Setup and testing protocol

The QZS isolation system under investigation consists in three independent legs (Figure 1a), having a vertical spring and three adjustable horizontal springs each, connected to the vertical guide by inclined bars. The stiffness of the horizontal and vertical springs are equal to 9.8 N/mm and 29.5 N/mm, respectively. A rigid steel plate, acting as payload, is then mounted on the three isolator legs by means of C-shaped steel beams. By adjusting the stiffness of the horizontal springs, the device exhibits a changing nonlinear dynamic behaviour. Figure 1b depicts an actual view of the equipment under test.

(a) (b)

Figure 1: (a) Single leg geometry; (b) Isolator under test and sensors positioning.

The device was instrumented in order to acquire both accelerations and displacements of some interest points. In particular, two tri-axial accelerometers have been mounted on the upper surface of the shaking table (R1) and on the center of the payload mass (A1), whereas three vertical uni-axial accelerometers have been deployed on the top plate of each leg (A2 to A4), as close as possible to the vertical guide in order to avoid to measure the effects of plate rotations. A picture of the whole experimental setup is reported in Figure 1b along with a view of the motion capture system used for the 3D tracking of the displacements of some critical points on the system.

To assess the dynamic features of the device and the related seismic protection level, several dynamic tests have been conducted on a 4 m $\times$ 4 m 6-DOF shaking table operating at the Experimental Dynamic Laboratory of the University of Enna "Kore" [4]. Two payload configurations and four different levels of the horizontal springs preload have been investigated. Particularly, payload configuration A and B refers to 501 kg and 624 kg, respectively, while the four preloads of the horizontal springs are indicated with the numbers 0, 20, 40 and 60 representing the shortening of the springs in mm. According to this, each set of tests can be identified by the combination of the payload and the preload, i.e. A-40.

Moreover, for each couple payload-preload, several dynamic tests have been conducted, for a total number of 235 tests, by exciting the system in vertical (z) direction with different input wave forms and amplitudes whose characteristics are:

- Random noise tests with frequency range from 0.25 Hz to 40 Hz and Root Mean Square (RMS) amplitude levels equals to 0.5 m/s^2 and 1.0 m/s^2;
- Continuous logarithmic sweep sine tests (upward and backward) with frequency range from 0.5 Hz to 16 Hz at different amplitude levels ranging from 0.1 m/s^2 to 1.0 m/s^2 according to the dynamic response of the system configuration;
- Stepped sine tests in the resonance frequency range, 0.5 Hz to 2.5 Hz, at different amplitude levels;
- Time history seismic vertical tests of the five selected natural earthquakes records listed in Table 1 and for four input levels (25%, 50% 75% and 100%).

The input earthquakes have been chosen for their different energy and frequency content to stress the system under several real seismic conditions.

Table 1: Natural earthquake time-histories applied.

Name	**Recording date**	**Recording station**	**z-axis PGA [m/s^2]**
Christchurch	21/02/2011	Heathcote Valley Primary School, Christchurch (NZ)	23.59
Bam	26/12/2003	Bam station, Kerman (Iran)	9.79
Chichi	20/09/1999	Stanion no. TCU084, Nantou county (Taiwan)	3.53
Gaziantep	06/02/2023	Pazarcik Kahramanmaras (Turkey)	5.85
Northridge	17/01/1994	Sylmar - Converter Sta, California (USA)	6.04

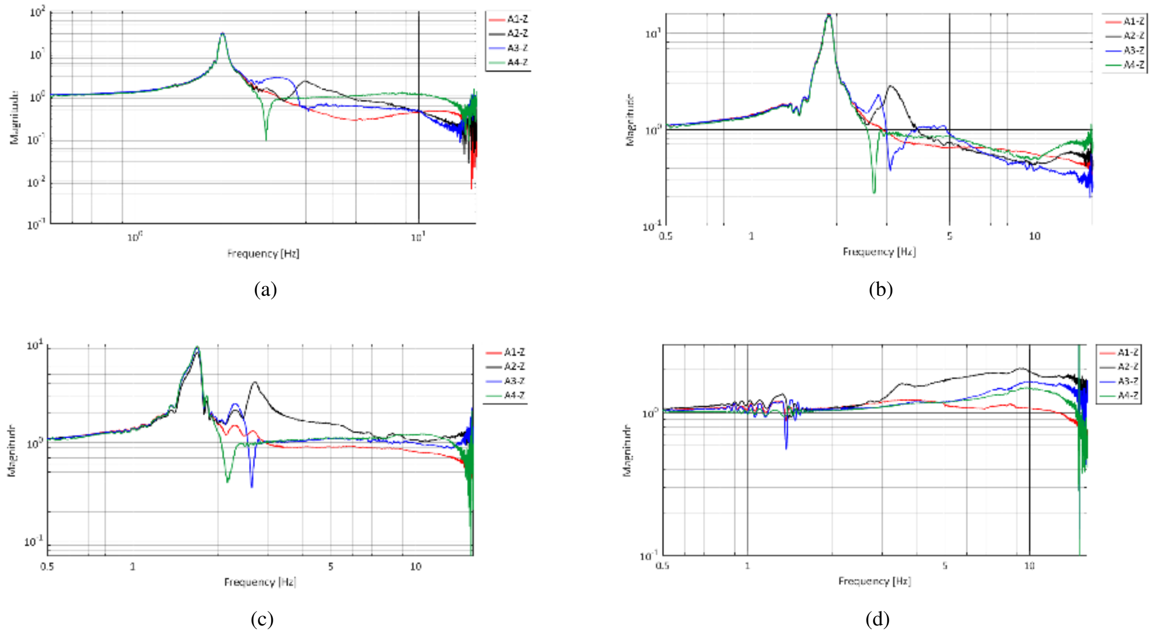
3 Experimental Results

All the experimental test have been executed with the aim to explore the dynamic properties the QZS system under several configuration of payload and stiffness preload and to assess its filtering properties with respect to the input acceleration. In this paper only a few experimental test results will be reported, for the sake of brevity.

Figure 2 depicts the Frequency Response Functions (FRF) magnitude of the output accelerometers A1Z-A4Z with respect to the shaking table acceleration R1Z computed during sweep sine test at 0.5 m/s^2, for the same payload (A) and different values of horizontal springs preload. As expected, the system in A-00 configuration exhibits a linear behaviour and the main resonance frequency observed is equal to 2.15 Hz while it is possible to observe a shift of the main resonance frequency to 1.80 Hz and 1.60 Hz for A-20 and A-40 configuration, respectively. It has been also observed that the dynamic behaviour become more complicated when rising the value of the springs preload and, in fact, several secondary peaks are evident for A-20 and A-40 while the FRF of the A-60 configuration does not exhibit a main resonance peak.

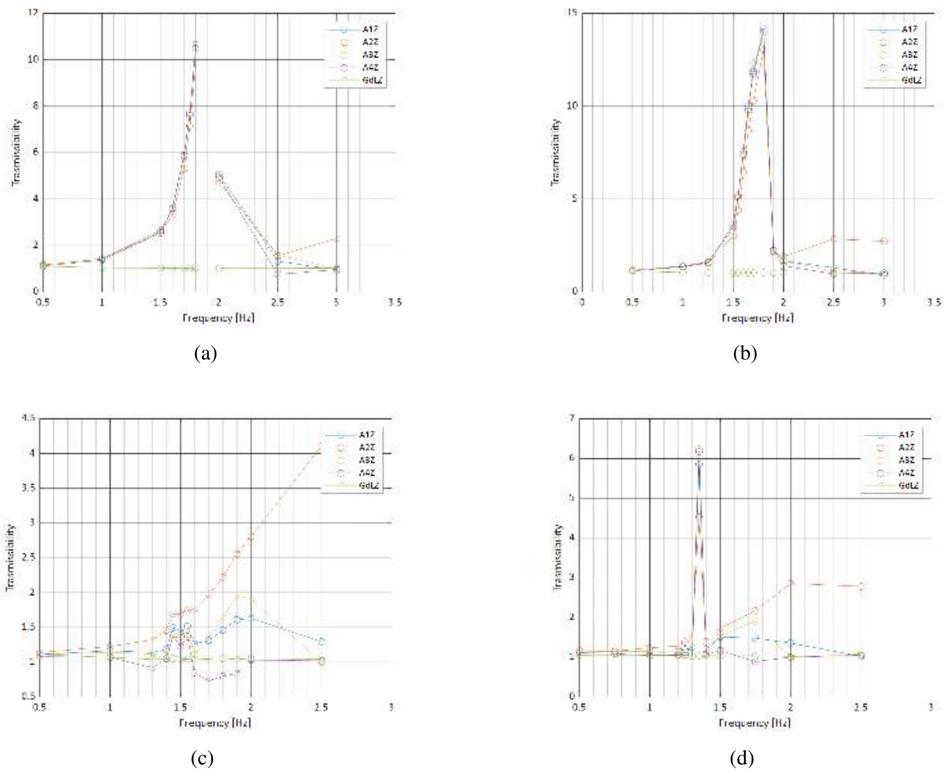
To further investigate the non-linear behaviour of the system a sequence of stepped sine tests have been conducted with frequency ranging between 0.5 Hz and 2.5 Hz, in order to focus on the resonance frequency, and by using different values of the sine wave amplitude. The analysis of Figure 3, where the transmissibility curves obtained for an input sine amplitude equal to 0.5 m/s^2 are reported, confirms that the system shows strong non-linearity even starting from low values of the horizontal springs preload (A-20 and A-40) where jumps in the transmissibility are observed close to the resonance frequency, while by increasing the preload (A-60) and the payload (B-40) the dynamic of the system is further altered and no evident resonance region is observed for the input amplitude of the test.

The primary objective of this study is to show the efficacy of the isolation system in mitigating the impact during seismic events. Figure 4 reports the trend of the peak values of all acceleration response channels (A1Z, A2Z, A3Z and A4Z) normalized with respect to the peak value of the input (R1Z) for the increasing values of the Christchurch earthquake amplitude. Similar results have been obtained for the other selected earthquakes but results are not reported here for brevity's sake. It is observed that the peck response is significantly attenuated



(a) (b) (c) (d)

Figure 2: Sweep sine test results: FRF magnitude for several values of the horizontal springs preload: (a) A-00; (b) A-20; (c) A-40; (d) A-60.



(a) (b) (c) (d)

Figure 3: Stepped sine test results: Transmissibility curves for several configurations of the system: (a) A-20; (b) A-40; (c) A-60; (d) B-40.

also for null (A-00) or low (A-40) values of the springs preload but in these cases the system is not able to properly attenuate the peak response for 100% seismic input. Best performances are obtained for higher values of the horizontal spring preload (A-60) and for higher values of the payload (B-40) in which case the isolation performances improve with the increase of the seismic input intensity and the peak reduction is higher than 60%.

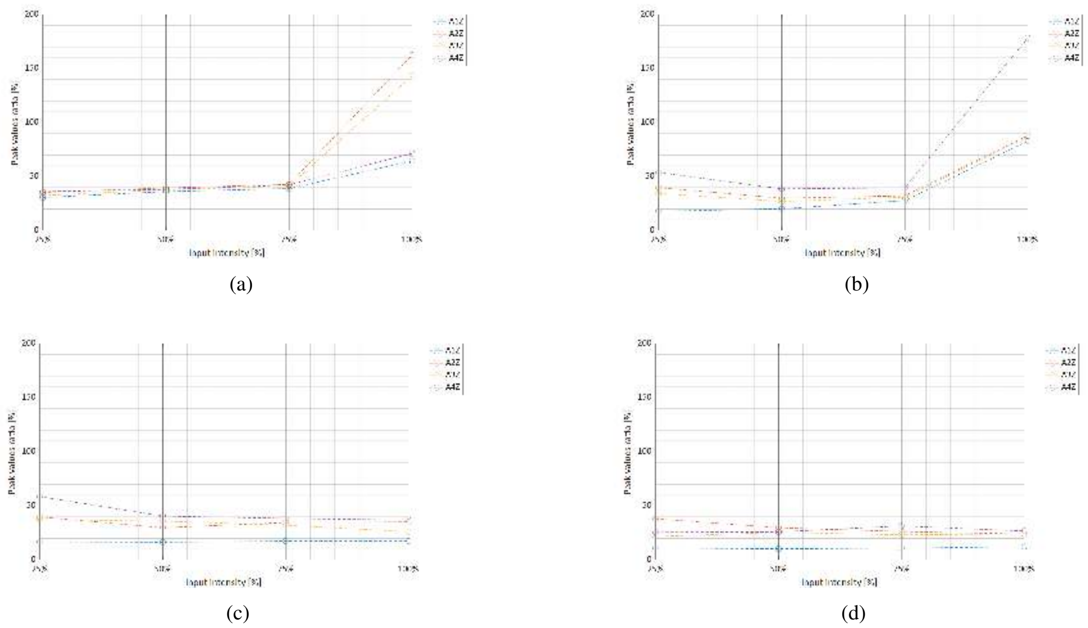


Figure 4: Normalized peak acceleration values of Christchurch earthquake for several configurations of the system: (a) A-00; (b) A-40; (c) A-60; (d) B-40.

Moreover, the results, particularly in terms of the Acceleration-Displacement hysteretic cycles (Figure 5), demonstrate the capability to filter and dissipate the input seismic energy. Indeed, very narrow and linear cycles are obtained for the A-00 configuration while an increase in the area of the cycles and non-linear backbone curves are observed as the preload on the horizontal springs is enhanced.

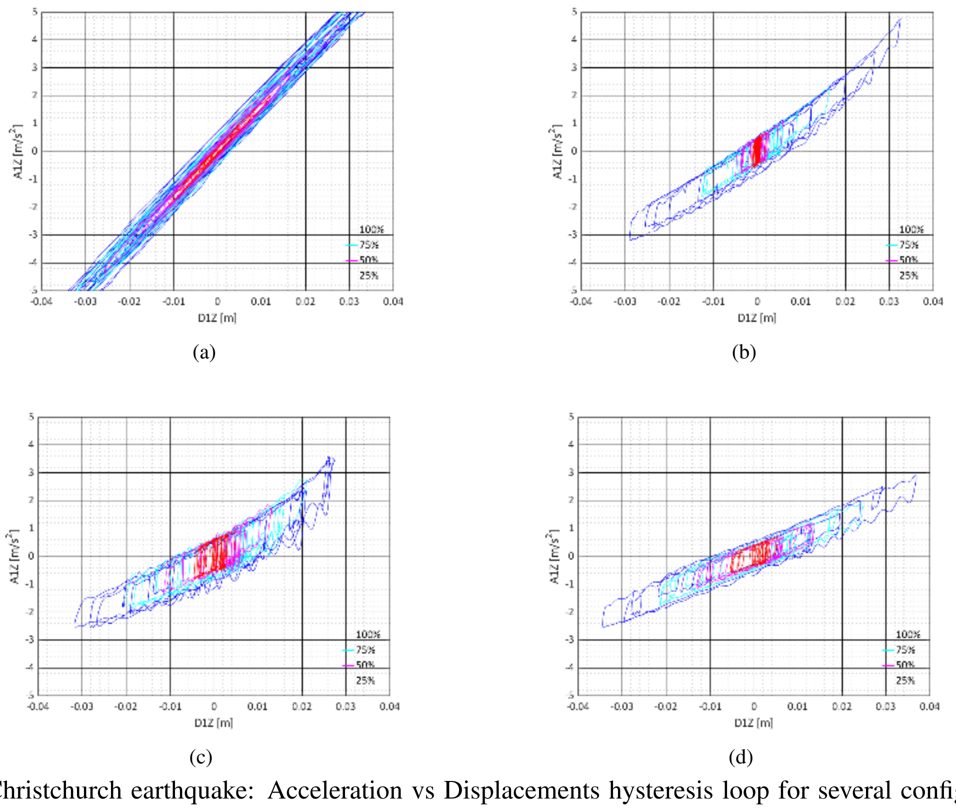


Figure 5: Christchurch earthquake: Acceleration vs Displacements hysteresis loop for several configurations of the system: (a) A-00; (b) A-40; (c) A-60; (d) B-40.

4 Conclusions

In this paper, the experimental campaign aimed at characterize and prove the effectiveness of a new High-Static-Low-Dynamic Stiffness Systems (HSLDSS) for vertical seismic isolation purposes is reported.

The experiments were conducted at the Experimental Dynamic Laboratory of the L.E.D.A. Research Institute, employing several input kind of input signals among which sweep sine, stepped sine, and natural earthquakes, to determine the dynamic properties of the isolation system, varying the payload mass and the horizontal stiffness preload.

The findings, in terms of acceleration and displacement of the isolated mass, affirm the efficacy of the proposed isolation system. Additionally, the device's robustness is evidenced by its proficient non-linear filtering performance in response to seismic inputs of varying amplitude, frequency content, and non-stationarity.

Future developments will focus on the numerical characterization of a non-linear analytical model able to simulate the actual dynamic behavior of the system for tuning and optimization purposes.

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